



Qualification Pack

Transformer Testing Supervisor

QP Code: CUTM/PSS/Q0101

Version: 1.0

NSQF Level: 5

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur
Uppalada || email:partha@cutm.ac.in



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CUTM/PSS/Q0101: Transformer Testing Supervisor

Brief Job Description

The individual is responsible to understand the testing requirement, transformer specification, verification of assembled transformer, testing functional and quality of transformer fabrication along with necessary safety norms.

Personal Attributes

A Transformer Testing Supervisor must have strong technical expertise in electrical testing, analytical thinking for fault diagnosis, and problem-solving skills for troubleshooting. They should possess leadership abilities to manage teams, attention to detail for quality assurance, safety awareness, effective communication, and time management to ensure transformers meet industry standards and customer requirements.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/PSS/N0101: Understand transformer testing requirements.](#)
2. [CUTM/PSS/N0102: Testing and quality check of assembled transformer](#)
3. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Power Distribution
Occupation	Transformer Manufacturing
Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NSQF

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Minimum Educational Qualification & Experience	Completed 3-year diploma (after 10th) (Mechanical / Electrical / Electronics / Production / Mechatronics Engineering) with 1 Year of experience OR Pursuing 2nd year of UG (Mechanical / Electrical / Electronics / Production / Mechatronics Engineering) with NA of experience OR Completed 2nd year of UG (UG Diploma) (Mechanical / Electrical / Electronics / Production / Mechatronics Engineering) with NA of experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-05-PW-00616-2023-V1-CUTM
NQR Version	1.0

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CUTM/PSS/N0101: Understand transformer testing requirements.

Description

This NOS covers the knowledge and skills required to understand transformer testing requirements for ensuring quality, performance, and compliance with standards. It includes interpreting technical specifications, understanding different types of transformers, and identifying key testing parameters such as insulation resistance, turns ratio, impedance, and load loss. The individual should be able to follow safety procedures, read testing reports, and comply with national and international standards. This competency ensures that transformer testing supervisors can effectively assess transformers' operational efficiency, diagnose faults, and maintain optimal performance in power distribution and industrial applications.

Scope

The scope covers the following :

- This NOS covers transformer testing requirements, including interpreting specifications, conducting insulation, impedance, and load loss tests, ensuring compliance with safety and industry standards, diagnosing faults, and analyzing test reports. It ensures transformers meet operational efficiency and quality standards in power distribution and industrial applications.

Elements and Performance Criteria

Identify Testing Parameters and Prepare Test Equipment

To be competent, the user/individual on the job must be able to:

- PC1.** Interact with Transformer Testing Engineer on customer requirement, engineering drawings and standards to be followed.
- PC2.** Understand tools, raw materials and consumables used for manufacturing.
- PC3.** Clearly understand the application of transformer and standards to be followed for testing process.
- PC4.** Understand detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific batch to be tested.
- PC5.** Understand the process steps to be followed during inspection.
- PC6.** Verify the work sheet steps on each stage of process followed matching to Client requirement.
- PC7.** Get necessary clarification from the appropriate authorities in case of any doubts.
- PC8.** Verify the working condition of all Testing equipment.
- PC9.** Check the calibration validity of the measuring instruments to be used for Testing.
- PC10.** Check incoming quality inspection reports on meeting the quality and quantity requirements.
- PC11.** Ensure tools required are placed in respective stages of transformer testing.
- PC12.** If tool is unavailable collect tools from stores, vendor site and place it in respective stages.
- PC13.** Update all testing details in appropriate logs, template, MIS.
- PC14.** Maintain test file, log, template, MIS and update as per requirement.

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- PC15.** Report to Head of manufacturing in case of any dispute in drawing, standards, materials, consumables, tools, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The basics of electricity, magnetism, and transformer working principles.
- KU2.** Knowledge of different transformer types such as power, distribution, instrument, and special-purpose transformers.
- KU3.** Transformer parts, including windings, core, insulation, oil, bushings, and cooling systems.
- KU4.** Knowledge of testing techniques like insulation resistance, turns ratio, short circuit, open circuit, and impulse voltage tests.
- KU5.** Common transformer faults (e.g., overheating, insulation failure, oil contamination) and their detection methods.
- KU6.** Awareness of electrical safety protocols, PPE requirements, and risk management during transformer testing.
- KU7.** Ability to document test results, prepare reports, and maintain compliance records.
- KU8.** Leadership, work delegation, and quality assurance in transformer testing environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate clearly with team members, engineers, and clients regarding transformer testing procedures and results.
- GS2.** Supervising and guiding the testing team to ensure efficiency and accuracy in operations.
- GS3.** Identify faults, analyzing issues, and making data-driven decisions for troubleshooting.
- GS4.** Efficiently planning and utilizing manpower, testing equipment, and other resources.
- GS5.** Follow workplace safety standards, electrical safety norms, and environmental regulations.
- GS6.** Expertise in various testing methods like insulation resistance, short circuit, open circuit, and oil testing.
- GS7.** Analyze test results to ensure compliance with industry standards and customer requirements.
- GS8.** Handle testing tools such as megger, TTR meter, high-voltage testers, and thermal cameras.
- GS9.** Maintain accurate records, preparing technical reports, and ensuring quality control in transformer testing processes.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify Testing Parameters and Prepare Test Equipment</i>	30	70	-	-
PC1. Interact with Transformer Testing Engineer on customer requirement, engineering drawings and standards to be followed.	2	6	-	-
PC2. Understand tools, raw materials and consumables used for manufacturing.	2	6	-	-
PC3. Clearly understand the application of transformer and standards to be followed for testing process.	2	6	-	-
PC4. Understand detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific batch to be tested.	2	5	-	-
PC5. Understand the process steps to be followed during inspection.	2	2	-	-
PC6. Verify the work sheet steps on each stage of process followed matching to Client requirement.	2	2	-	-
PC7. Get necessary clarification from the appropriate authorities in case of any doubts.	2	4	-	-
PC8. Verify the working condition of all Testing equipment.	2	3	-	-
PC9. Check the calibration validity of the measuring instruments to be used for Testing.	2	6	-	-
PC10. Check incoming quality inspection reports on meeting the quality and quantity requirements.	2	6	-	-
PC11. Ensure tools required are placed in respective stages of transformer testing.	2	6	-	-
PC12. If tool is unavailable collect tools from stores, vendor site and place it in respective stages.	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Update all testing details in appropriate logs, template, MIS.	2	6	-	-
PC14. Maintain test file, log, template, MIS and update as per requirement.	2	3	-	-
PC15. Report to Head of manufacturing in case of any dispute in drawing, standards, materials, consumables, tools, etc.	2	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0101
NOS Name	Understand transformer testing requirements.
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

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CUTM/PSS/N0102: Testing and quality check of assembled transformer

Description

This NOS covers testing and quality checks of assembled transformers to ensure performance, safety, and compliance with standards. It includes conducting electrical tests, verifying insulation, assessing load performance, identifying defects, and analyzing test reports to certify transformer reliability in power distribution and industrial applications.

Scope

The scope covers the following :

- This NOS includes conducting electrical tests, insulation checks, load performance assessments, and verifying compliance with standards. It covers fault detection, quality assurance, report analysis, and ensuring transformer reliability. It ensures assembled transformers meet operational, safety, and industry standards for efficient power distribution and industrial applications.

Elements and Performance Criteria

Various testing methods as per transformer design and applications

To be competent, the user/individual on the job must be able to:

- PC1.** understand various testing methods to be followed as per design and application.
- PC2.** prepare check list of parameters for consideration during testing and inspection.
- PC3.** verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections.
- PC4.** perform routine test, loss measurement and temperature rise test, dielectric test.
- PC5.** carry out insulation-resistance test to identify grounding and shorting faults in the connections.
- PC6.**
 - check for any damage, flash, and hair crack with respect to specified parameters.
 - Parameters: condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket joints and flanges
- PC7.** inspect excess humming noise in transformer and identify loose fitting of silicon mixed steel alloys laminated core joints.

Quality check and confirmation of proper assembly

To be competent, the user/individual on the job must be able to:

- PC8.**
 - identify faults by visually inspecting general appearance, leakage of oil, and physical condition
 - Physical condition: rust on body and radiators
- PC9.** check silica gel breather for the oil level in oil cap on specific parameters.
- PC10.**
 - identify faults
 - Faults: faults due to primary winding, braze /solder of LT winding joints melted, overheat, open circuit in internal wiring etc.
- PC11.** to maintain records of transformer Basic information: serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the transformer performance history.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** The transformer principles, electromagnetic induction, and types of transformers.
- KU2.** Various transformer parts such as core, windings, insulation, bushings, tap changers, and cooling systems.
- KU3.** the importance of visual inspection, insulation checks, and ensuring proper assembly before testing.
- KU4.** The key transformer tests like turns ratio test, insulation resistance test, dielectric strength test, and winding resistance test.
- KU5.** different types of defects such as insulation failure, short circuits, overheating, and improper connections.
- KU6.** safety measures, PPE usage, grounding techniques, and emergency handling in transformer testing.
- KU7.** to record test results, prepare quality check reports, and ensure compliance with specifications.
- KU8.** customer requirements, efficiency targets, and industry quality standards before dispatch.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** clearly explain test results, quality standards, and troubleshooting steps to team members and supervisors.
- GS2.** Identify transformer defects, analyzing test data, and determining corrective actions.
- GS3.** Ensure accuracy in test procedures, data recording, and compliance with quality standards.
- GS4.** follow electrical safety guidelines, PPE usage, and emergency protocols in the testing environment.
- GS5.** test insulation resistance, turns ratio, short circuit, and oil breakdown voltage tests.
- GS6.** Analyze and validate test results against industry standards and manufacturer specifications.
- GS7.** handle megger, TTR meter, high-voltage testers, and infrared thermography tools.
- GS8.** Prepare quality check reports, maintaining records, and documenting non-conformities for corrective action.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Various testing methods as per transformer design and applications</i>	26	39	-	-
PC1. understand various testing methods to be followed as per design and application.	4	6	-	-
PC2. prepare check list of parameters for consideration during testing and inspection.	4	6	-	-
PC3. verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections.	2	3	-	-
PC4. perform routine test, loss measurement and temperature rise test, dielectric test.	4	6	-	-
PC5. carry out insulation-resistance test to identify grounding and shorting faults in the connections.	4	6	-	-
PC6. • check for any damage, flash, and hair crack with respect to specified parameters. • Parameters: condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket joints and flanges	4	6	-	-
PC7. inspect excess humming noise in transformer and identify loose fitting of silicon mixed steel alloys laminated core joints.	4	6	-	-
<i>Quality check and confirmation of proper assembly</i>	14	21	-	-
PC8. • identify faults by visually inspecting general appearance, leakage of oil, and physical condition • Physical condition: rust on body and radiators	4	6	-	-
PC9. check silica gel breather for the oil level in oil cap on specific parameters.	2	3	-	-
PC10. • identify faults • Faults: faults due to primary winding, braze /solder of LT winding joints melted, overheat, open circuit in internal wiring etc.	4	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. to maintain records of transformer Basic information: serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the transformer performance history.	4	6	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0102
NOS Name	Testing and quality check of assembled transformer
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

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CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Detailed Guidelines for Assessment – Transformer Testing Supervisor (NSQF Level 5)

Assessment Mode: Theory, Practical, and Viva-Based Evaluation

The assessment for this qualification will be conducted in two components:

Theory Examination (40%) – Written MCQs and descriptive questions.

Practical Assessment (60%) – Hands-on evaluation in a lab/testing environment.

1. General Assessment Strategy

- The assessment should take place in a controlled environment, such as a transformer testing lab or training facility.
- Batches will be assigned to Assessment Agencies, which will deploy ToA-certified Assessors to conduct the assessment.



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- The minimum passing percentage is 70% overall and 70% in each NOS/module.
- Surprise audits and video call verification will be conducted to ensure quality assurance.

2. Assessment for Each NOS

NOS 1: Understand Transformer Testing Requirements

Assessment Components:

- Theory Exam (30 marks) – Covers topics like testing standards, transformer specifications, tools, and raw materials.
- Practical Exam (70 marks) – Includes verification of test equipment, calibration checks, and documentation procedures.

NOS 2: Testing and Quality Check of Assembled Transformer

Assessment Components:

- Theory Exam (40 marks) – Includes transformer testing methods, inspection checklists, fault identification, and quality control.
- Practical Exam (60 marks) – Hands-on testing, measuring electrical parameters, and analyzing defects.

NOS 3: Ensure Organizational Health, Safety, and Security

Assessment Components:

- Theory Exam (30 marks) – Covers hazard identification, safety procedures, and emergency protocols.
- Practical Exam (70 marks) – Demonstration of PPE usage, risk mitigation, and workplace safety compliance.

NOS 4: Employability Skills

Assessment Components:

- Theory Exam (30 marks) – Includes career development, digital skills, communication, and financial literacy.
- Practical Exam (70 marks) – Role plays, case studies, and presentations on workplace ethics and teamwork.

3. Assessment Conditions

- Qualified Assessors: B.Tech in Electrical/Mechanical/Electronics with 3 years of industry experience and 1 year of training experience.
- Lead Assessors & Proctors: Must have at least 7 years of industry experience.
- Assessment Location: Conducted in a transformer testing facility or training center with access to

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equipment and testing tools.

- Assessment Tools & Equipment: Includes insulation testers, power analyzers, winding resistance meters, and digital multimeters.

- Mode of Assessment: Offline

4. Assessment Documentation:

- Geo-tagged photos of assessments.
- Video recordings for monitoring and auditing.
- Hard copies and soft copies of assessment records.

5. Passing Criteria & Reassessment

- Minimum Pass Percentage: 70% in aggregate and 70% in each NOS/module.
- Candidates failing the assessment will be given one reassessment opportunity after additional training.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/PSS/N0101.Understand transformer testing requirements.	30	70	0	0	100	30
CUTM/PSS/N0102.Testing and quality check of assembled transformer	40	60	0	0	100	30
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	0	0	100	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	20



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	120	230	-	-	350	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.